Summary of Radiation Field by a Moving Charge

Total electromagnetic field emitted by a moving charge

$$\begin{aligned}\vec{E}(\vec{x},t) &= q\left\{\frac{\vec{n}-\vec{\beta}}{\gamma^2 R^2 (1-\vec{n}\cdot\vec{\beta}\,)^3} + \frac{\vec{n}\times[(\vec{n}-\vec{\beta}\,)\times\dot{\vec{\beta}}\,]}{c\,R\,(1-\vec{n}\cdot\vec{\beta}\,)^3}\right\}_{retard} \\ \vec{B}(\vec{x},t) &= \left(\vec{n}\times\vec{E}\right)_{retard}\end{aligned}$$

where the retardation condition is $ct = ct_r + R = ct_r + \vec{n}\cdot(\vec{x}-\vec{r}\,)$. The radiation field is

$$\vec{E}_{rad}(\vec{x},t) = q\,\frac{\vec{n}\times[(\vec{n}-\vec{\beta}\,)\times\dot{\vec{\beta}}\,]}{c\,R\,(1-\vec{n}\cdot\vec{\beta}\,)^3}\bigg|_{retard} \qquad \text{and} \qquad \vec{B}_{rad}(\vec{x},t) = \vec{n}\times\vec{E}_{rad}(\vec{x},t)$$

Polarization of the Field

$\vec{B}$ is always perpendicular to the direction of the propagation direction $\vec{n}$, *i.e.* $\vec{B}\cdot\vec{n}=0$. The static part of $\vec{E}$ has a component in the direction of $\vec{n}$, *i.e.*

$$\vec{n}\cdot\vec{E} = \frac{q}{\gamma^2 R^2 (1-\vec{n}\cdot\vec{\beta}\,)^2}$$

while $\vec{E}_{rad}$ is perpendicular to both $\vec{n}$ and $\vec{B}$, $\vec{E}_{rad}\cdot\vec{n}=0$ and $\vec{B}\cdot\vec{E}=0$.

Radiation Power Per Solid Angle

$$\left(\frac{dP}{d\Omega}\right)_{rad} = \frac{c}{4\pi}R^2|\vec{E}_{rad}|^2 = \frac{c}{4\pi}R^2|\vec{B}_{rad}|^2$$

Note that the energy radiated through a unit solid angle is independent of R, the distance to the observer, *i.e.* the total radiation energy on a sphere that is centered at the radiation source does not decay with the radius R.

Frequency Spectrum of Radiation Energy

$$\left(\frac{d^2E}{d\Omega d\omega}\right)_{rad} = \frac{1}{\pi}\left|\vec{A}(\omega)\right|^2 = \frac{1}{\pi}\left|\vec{C}(\omega)\right|^2$$

where

$$\vec{A}(\omega) = \sqrt{\frac{c}{4\pi}}\int_{-\infty}^{\infty} R\vec{E}_{rad}(\vec{r},t)\,e^{i\omega t}\,dt$$

and

$$\vec{C}(\omega) = \sqrt{\frac{c}{4\pi}}\int_{-\infty}^{\infty} R\vec{B}_{rad}(\vec{r},t)\,e^{i\omega t}\,dt = \sqrt{\frac{c}{4\pi}}\int_{-\infty}^{\infty} R\left[\vec{n}\times\vec{E}_{rad}(\vec{r},t)\right]\,e^{i\omega t}\,dt$$

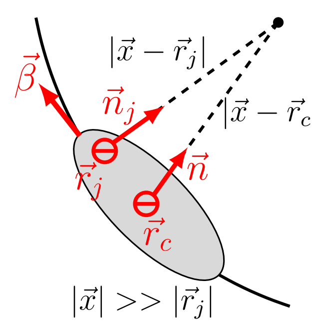
For $\gamma >> 1$, the radiation is concentrated inside a very narrow forward radiation cone with $\Delta\theta \sim 1/\gamma$, $d\vec{n}/dt \simeq 0$ and

$$\begin{aligned}
\vec{A}(\omega) &= \frac{q}{\sqrt{4\pi c}}\, e^{i\omega(\vec{x}\cdot\vec{n})/c} \int_{-\infty}^{\infty} \frac{\vec{n}\times[(\vec{n}-\vec{\beta}\,)\times\dot{\vec{\beta}}\,]}{(1-\vec{n}\cdot\vec{\beta}\,)^2}\, e^{i\omega[t-\vec{n}\cdot\vec{r}(t)/c]}\, dt \\
&= -\frac{i\omega q}{\sqrt{4\pi c}}\, e^{i\omega(\vec{x}\cdot\vec{n})/c} \int_{-\infty}^{\infty} [\vec{n}\times(\vec{n}\times\vec{\beta}\,)]\, e^{i\omega[t-\vec{n}\cdot\vec{r}(t)/c]}\, dt \qquad (1) \\
\vec{C}(\omega) &= \sqrt{\frac{c}{4\pi}} \int_{-\infty}^{\infty} R\left[\vec{n}\times\vec{E}_{rad}(\vec{r},t)\right] e^{i\omega t}\, dt \;=\; \vec{n}\times\left[\sqrt{\frac{c}{4\pi}} \int_{-\infty}^{\infty} R\vec{E}_{rad}(\vec{r},t)\, e^{i\omega t}\, dt\right] \\
&= \vec{n}\times\vec{A}(\omega) \;=\; \frac{i\omega q}{\sqrt{4\pi c}}\, e^{i\omega(\vec{x}\cdot\vec{n})/c} \int_{-\infty}^{\infty} (\vec{n}\times\vec{\beta}\,)\, e^{i\omega[t-\vec{n}\cdot\vec{r}(t)/c]} dt \qquad (2)
\end{aligned}$$

where $\vec{n}\times[\vec{n}\times(\vec{n}\times\vec{\beta}\,)] = (\vec{n}\times\vec{\beta}\,)$. Note that, in Eqs. (1) and (2), there is no explicit dependence of $\dot{\vec{\beta}}$ and it seems like a moving particle without acceleration can also radiate electromagnetic field. This is just a false impression. The consideration of the acceleration is included with the trajectory $\vec{r}(t)$ of the particle in Eq. (1) and (2).

Chapter 17. More On Radiation By Moving Charged Particles

17.1 Coherent and Incoherent Radiation From a Beam of Charged Particles



The Fourier transformation of radiation B-field from a charged particle in Eq. (2) can be written as

$$\vec{C}(\omega) \;=\; iq\sqrt{\frac{c}{4\pi}}\, e^{i\vec{k}\cdot\vec{x}} \int_{-\infty}^{\infty} (\vec{k}\times\vec{\beta})\, e^{i[\omega t-\vec{k}\cdot\vec{r}(t)]} dt$$

where $\vec{k} = (\omega/c)\vec{n}$ is the wave vector of the radiation. If there are N identical charged particles in a particle bunch, the radiation field is

$$\vec{C}(\omega) = iq\sqrt{\frac{c}{4\pi}} \sum_{j=1}^{N} \left[e^{i\omega(\vec{x}\cdot\vec{n}_j)/c} \int_{-\infty}^{\infty} (\vec{k}_j\times\vec{\beta}_j)\, e^{i[\omega t-\vec{k}_j\cdot\vec{r}_j(t)]} dt\right] \qquad (3)$$

where $\vec{r}_j(t)$ is the location of particle j at time t and $\vec{n}_j = [\vec{x}-\vec{r}_j(t)]/|\vec{x}-\vec{r}_j(t)|$. The radiation energy from a bunch of charged particles is then

$$\left(\frac{d^2E}{d\omega d\Omega}\right)_N \;=\; \frac{1}{\pi}\left|\vec{C}(\omega)\right|^2 \;=\; \frac{q^2 c}{4\pi^2}\left|\sum_{j=1}^{N} e^{i\omega(\vec{x}\cdot\vec{n}_j)/c} \int_{-\infty}^{\infty} (\vec{k}\times\vec{\beta}_j)\, e^{i[\omega t-\vec{k}\cdot\vec{r}_j(t)]}\, dt\right|^2 \qquad (4)$$

Since $|\vec{x}| >> |\vec{r}_j|$, $\vec{n}_j \simeq [\vec{x}-\vec{r}_c]/|\vec{x}-\vec{r}_c| = \vec{n}$, $\vec{k}_j = (\omega/c)\vec{n}_j \simeq \vec{k}$, where $\vec{r}_c$ is the center of the particle bunch and $d\vec{n}/dt \simeq 0$, the radiation energy from a bunch of charged particles can be calculated from

$$\begin{aligned}
\left(\frac{d^2E}{d\omega d\Omega}\right)_N &= \frac{q^2c}{4\pi^2}\left|\sum_{j=1}^{N}\int_{-\infty}^{\infty}(\vec{k}\times\vec{\beta}_j)\, e^{i[\omega t-\vec{k}\cdot\vec{r}_j(t)]}\,dt\right|^2 \\
&= \frac{q^2c}{4\pi^2}\iint_{-\infty}^{\infty} dt_1dt_2\, e^{i\omega(t_1-t_2)}\sum_{i=1}^{N}\sum_{j=1}^{N}[\vec{k}\times\vec{\beta}_j(t_1)]\cdot[\vec{k}\times\vec{\beta}_i(t_2)]\, e^{i\vec{k}\cdot[\vec{r}_i(t_2)-\vec{r}_j(t_1)]} \\
&= \frac{q^2c}{4\pi^2}\sum_{j=1}^{N}\left|\int_{-\infty}^{\infty}(\vec{k}\times\vec{\beta}_j)\, e^{i\left[\omega t-\vec{k}\cdot\vec{r}_j(t)\right]}dt\right|^2 \\
&+ \frac{q^2c}{4\pi^2}\iint_{-\infty}^{\infty} dt_1dt_2\, e^{i\omega(t_1-t_2)}\sum_{i\neq j}^{N}\sum_{j=1}^{N}[\vec{k}\times\vec{\beta}_j(t_1)]\cdot[\vec{k}\times\vec{\beta}_i(t_2)]\, e^{i\vec{k}\cdot[\vec{r}_i(t_2)-\vec{r}_j(t_1)]} \qquad (5)
\end{aligned}$$

where

$$[\vec{k}\times\vec{\beta}_j(t_1)]\cdot[\vec{k}\times\vec{\beta}_i(t_2)] = k^2[\vec{\beta}_j(t_1)\cdot\vec{\beta}_i(t_2)] - [\vec{k}\cdot\vec{\beta}_j(t_1)][\vec{k}\cdot\vec{\beta}_i(t_2)]$$

In Eq. (5), the first term with a single summation of the particles is the incoherent sum of the radiation intensity from individual particles while the second term with double summations over the particles is the interference between the radiation fields from different particles.

Superposition of the Radiation Field from N Sources

Consider the total electric field from N similar radiation sources with different phase arriving at an observation point, the energy intensity of the radiation is

$$I_{tot} \sim \left|\sum_{i=1}^{N}\vec{E}_i\, e^{i\phi_i}\right|^2 = \sum_{i=1}^{N}E_i^2 + \sum_{i\neq j}^{N}\sum_{j=1}^{N}2\vec{E}_i\cdot\vec{E}_j\cos(\phi_i-\phi_j)$$

where $E_1 \simeq E_2 \simeq \cdots \simeq E_N$ for simplicity. The first term with single summation is the incoherent addition of the radiation and the second term with $\cos(\phi_i-\phi_j)$ is the interference of the fields from N different sources. The coherent radiation observed at the observation point occurs when the radiation from difference sources arrives at the observation point with a common phase, *i.e.*

Coherent Radiation: $\quad \phi_1 = \phi_2 = \cdots = \phi_N \quad \Longrightarrow \quad \cos(\phi_i-\phi_j) = 1$

$$\Longrightarrow \quad I_{tot} = \sum_{i=1}^{N}E_i^2 + \sum_{i\neq j}^{N}\sum_{j=1}^{N}2\vec{E}_i\cdot\vec{E}_j = \left|\sum_{i=1}^{N}\vec{E}_i\right|^2 = N^2I_1$$

where $I_1 \sim |\vec{E}_1|^2$ is the radiation intensity from a single source. For incoherent radiation, the phases of the radiation field from N different sources arriving at the observation point are all different randomly, *i.e.*

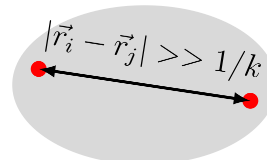
Incoherent Radiation: $\quad \phi_1 \neq \phi_2 \neq \cdots \neq \phi_N$

$$\Longrightarrow \quad \sum_{i\neq j}^{N}\sum_{j=1}^{N}\cos(\phi_i-\phi_j) \overset{N\to\infty}{=} \int_0^{2\pi}\cos\delta\, d\delta = 0 \quad \Longrightarrow \quad I_{tot} = \sum_{i=1}^{N}E_i^2 = NI_1$$

Therefore, the coherent radiation from N radiation sources is N times stronger than the incoherent radiation. The coherent radiation is thus much stronger than the incoherent radiation when $N \longrightarrow \infty$.

17.1.1 Incoherent Radiation

If the motion of N particles in a bunch are uncorrelated and the positions of particles are sufficiently spread so that the bunch size is much larger than the wave length of the radiation, *i.e.*



$$\max(\vec{k}\cdot(\vec{r}_i - \vec{r}_j)) >> 1 \qquad \forall\,(i,j)$$

For $N \longrightarrow \infty$, the oscillation phase of $e^{i\vec{k}\cdot[\vec{r}_i(t_2)-\vec{r}_j(t_1)]}$ of the interference term in Eq. (5) is uniformly distributed in $[0,\,2\pi)$ and the interferences among the radiation from different particles cancel each other as

$$\sum_{i\neq j}^{N}[\vec{k}\times\vec{\beta}_j(t_1)]\cdot[\vec{k}\times\vec{\beta}_i(t_2)]\,e^{i\vec{k}\cdot[\vec{r}_i(t_2)-\vec{r}_j(t_1)]}$$
$$\simeq\ [\vec{k}\times\vec{\beta}(t_1)]\cdot[\vec{k}\times\vec{\beta}(t_2)]\sum_{i\neq j}^{N}e^{i\vec{k}\cdot[\vec{r}_i(t_2)-\vec{r}_j(t_1)]}\ =\ 0$$

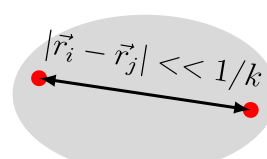
When the bunch size is larger than the wavelength of the radiation, therefore, the intensity of the radiation from N particles is the sum of the radiation intensity of individual particles,

$$\left(\frac{d^2E}{d\omega d\Omega}\right)_N \ \simeq\ \frac{q^2c}{4\pi^2}\sum_{j=1}^{N}\left|\int_{-\infty}^{\infty}(\vec{k}\times\vec{\beta}_j)\,e^{i\left[\omega t-\vec{k}\cdot\vec{r}_j(t)\right]}dt\right|^2 \ =\ N\left(\frac{d^2E}{d\omega d\Omega}\right)_1 \tag{6}$$

where the intensity of the incoherent radiation by N particles is N times of the radiation intensity by a single particle.

17.1.2 Coherent Radiation

If the bunch size is sufficiently small, the motion of the particles within a bunch is negligible as compared with the motion of the bunch. The radiations from the bunch particles are then added constructively and the intensity of the radiation from N particles will be N^2 times of the radiation intensity by a single particle. The condition for the constructive superposition of the radiation field in Eq. (5) is the bunch size being much smaller than the wave length of the radiation so that



$$\vec{k}\cdot(\vec{r}_i - \vec{r}_j) << 1 \qquad \forall\,(i,j) \tag{7}$$

For a charged-particle beam in an accelerator, a particle bunch moves at a CM velocity close to the speed of light and the motion of bunch particles relative to the CM is negligible as compared with the CM velocity. The position of particles within a bunch is random and uniformly distributed. If the bunch size satisfies the coherency condition in Eq. (7). the trajectory of each particle within a bunch can be approximated as

$$\vec{r}_j(t) = \vec{\xi}_j + \vec{r}(t) = \vec{\xi}_j + \int \vec{v}(t)\, dt$$

where $\vec{r}(t)$ is the CM location, $\vec{\xi}_j$ is the position of particle j relative to the CM, and $\vec{\xi}_j$ is approximately constant during the pulse of the radiation within the radiation cone pointing toward an observer at $\vec{X}$. The coherency condition in Eq. (7) can be expressed as

$$\vec{k} \cdot (\vec{r}_i - \vec{r}_j) = \vec{k} \cdot (\vec{\xi}_i - \vec{\xi}_j) << 1 \qquad \Longrightarrow \qquad \vec{k} \cdot \vec{\xi}_i << 1 \tag{8}$$

The radiation intensity from a bunch of particles is then

$$\begin{aligned} \left(\frac{d^2E}{d\omega d\Omega}\right)_N &= \frac{q^2 c}{4\pi^2} \left| \sum_{j=1}^{N} \int_{-\infty}^{\infty} (\vec{k} \times \vec{\beta}_j)\, e^{i[\omega t - \vec{k}\cdot\vec{r}_j(t)]}\, dt \right|^2 \\ &= \frac{q^2}{4\pi^2 c} \left| \left[\int_{-\infty}^{\infty} (\vec{k} \times \vec{v}\,)\, e^{i[\omega t - \vec{k}\cdot\vec{r}(t)]}\, dt \right] \sum_{j=1}^{N} e^{-i\vec{k}\cdot\vec{\xi}_j} \right|^2 \\ &= \left(\frac{d^2E}{d\omega d\Omega}\right)_1 \left| \sum_{j=1}^{N} e^{-i\vec{k}\cdot\vec{\xi}_j} \right|^2 \end{aligned} \tag{9}$$

where

$$\left(\frac{d^2E}{d\omega d\Omega}\right)_1 = \frac{q^2}{4\pi^2 c} \left| \int_{-\infty}^{\infty} (\vec{k} \times \vec{v}\,)\, e^{i[\omega t - \vec{k}\cdot\vec{r}(t)]}\, dt \right|^2$$

is the intensity of the radiation of a single particle traveling at the CM of a bunch. With $\vec{k} \cdot \vec{\xi}_j << 1$,

$$\left| \sum_{j=1}^{N} e^{-i\vec{k}\cdot\vec{\xi}_j} \right|^2 \simeq \left| \sum_{i=1}^{N} 1 \right|^2 = N^2$$

Therefore, the intensity of the coherent radiation is

$$\left(\frac{d^2E}{d\omega d\Omega}\right)_N \simeq N^2 \left(\frac{d^2E}{d\omega d\Omega}\right)_1$$

In general, we can define a form factor of a bunch

$$\left| \sum_{i=1}^{N} e^{-i\vec{k}\cdot\vec{\xi}_i} \right|^2 = N^2 F(\vec{k}\,) \qquad \text{where} \quad 0 < F(\vec{k}\,) \leq 1 \tag{10}$$

and then

$$\left(\frac{d^2E}{d\omega d\Omega}\right)_N = N^2 F(\vec{k}) \left(\frac{d^2E}{d\omega d\Omega}\right)_1 \tag{11}$$

The form factor of a bunch can be calculated from the particle distribution (charge density) $\rho(\vec{\xi}\,)$ of a bunch as

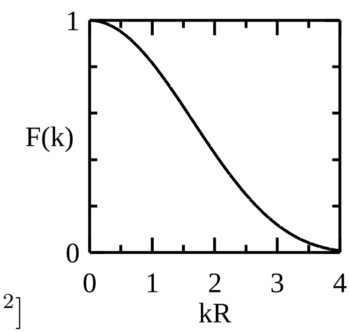
$$F(\vec{k}\,) = \left|\frac{1}{N}\sum_{i=1}^{N} e^{-i\vec{k}\cdot\vec{\xi}_i}\right|^2 = \left|\frac{1}{N}\int e^{-i\vec{k}\cdot\vec{\xi}}\rho(\vec{\xi}\,)d\vec{\xi}\right|^2$$

which is the Fourier transformation of the charge density of a bunch.

Examples of Form Factor $F(\vec{k}\,)$ of a Bunch of Charged Particles

a) N particles are uniformly distributed inside a sphere of radius R.

$$\begin{aligned}\int e^{-i\vec{k}\cdot\vec{r}}\rho(\vec{r}\,)\,d\vec{r} &= \frac{3N}{4\pi R^3}\int_0^{2\pi}\int_0^{\pi}\int_0^{R} e^{-ikr\cos\theta}r^2\sin\theta\,drd\theta d\phi = \frac{3N}{ikR^3}\int_0^{R}\sin(kr)rdr \\ &= \frac{3N}{ik^3R^3}\left[\sin(kR)-kR\cos(kR)\right]\end{aligned}$$



The form factor is

$$F(k) = \frac{9}{k^6R^6}\left[\sin(kR)-kR\cos(kR)\right]^2$$

When $kR << 1$,

$$\begin{aligned}\sin(kR)-kR\cos(kR) &\simeq kR-\frac{1}{6}(kR)^3-kR[1-\frac{1}{2}(kR)^2] \\ &= (kR)^3/3\end{aligned}$$

Therefore, $F \simeq 1$ when $kR << 1$. When $kR >> 1$ or $\lambda << R\,,\ F \sim 1/(kR)^4 \longrightarrow 0\,.$

b) N particles in a Gaussian distribution

$$\rho(\vec{r}\,) = \frac{N}{(2\pi\sigma^2)^{3/2}}e^{-r^2/(2\sigma^2)}$$

Choose a coordinate so that $\vec{k}$ is along $\vec{e}_x$ and then

$$\begin{aligned}\int e^{-i\vec{k}\cdot\vec{r}}\rho(\vec{r}\,)\,d\vec{r} &= \frac{N}{(2\pi\sigma^2)^{3/2}}\iiint_{-\infty}^{\infty} e^{-ikx}e^{-(x^2+y^2+z^2)/(2\sigma^2)}dxdydz \\ &= \frac{N}{(2\pi\sigma^2)^{1/2}}\int_{-\infty}^{\infty} e^{-ikx-x^2/(2\sigma^2)}dx = Ne^{-\sigma^2k^2/2}\end{aligned}$$

and

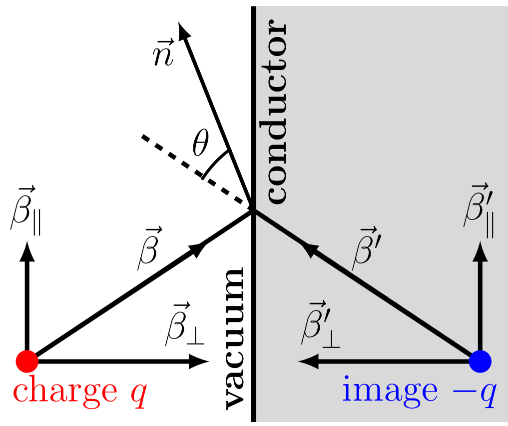
$$F(k) = e^{-\sigma^2k^2}$$

which decays to zero exponentially when $k = \omega/c > 1/\sigma$ or $\lambda < \sigma$ and $F \simeq 1$ for $\sigma << \lambda$.

In both examples, the radiation spectrum falls off to zero sharply when the wavelength is smaller than the size of the particle bunch. This provides a way to measure the size of a charged-particle bunch experimentally.

17.2 Transition Radiation (Section 13.7 in Jackson's Book)

When a charged particle collides with a conductor, it is absorbed by the conductor. The radiation emitted upon the abrupt disappearance of the charge is called transition radiation. For a perfect conductor, the interaction between a charged particle and a conductor can be treated by the interaction between the charge and its image charge. When the charge is absorbed by a conductor, the charge and its image charge are cancelled out at the surface of the conductor. In this case, for a easier calculation, the frequency spectrum of the radiation is calculated using the $\vec{B}$-field



$$\beta = \beta', \quad \vec{\beta}_{\perp} = -\vec{\beta}'_{\perp}, \quad \vec{\beta}_{\parallel} = \vec{\beta}'_{\parallel}$$

$$\vec{\beta} \simeq \text{constant}$$

$$\frac{d^2E}{d\omega d\Omega} = \frac{1}{\pi}\left|\vec{C}(\omega)\right|^2$$

where $\vec{C}(\omega)$ is the sum of the radiation fields from the charge and its image charge,

$$\vec{C}(\omega) = \frac{i\omega q}{\sqrt{4\pi c}}\, e^{i\omega(\vec{x}\cdot\vec{n})/c} \int_{-\infty}^{0} dt \left\{ \left(\vec{n}\times\vec{\beta}\right) e^{i\omega[t-\vec{n}\cdot\vec{r}(t)/c]} - \left(\vec{n}\times\vec{\beta}\,'\right) e^{i\omega[t-\vec{n}\cdot\vec{r}\,'(t)/c]} \right\} \tag{12}$$

where the first and second term in Eq. (12) are the radiation field of the charge and its image charge, respectively, and the charge collides the conductor surface at $t = 0$. For $t > 0$, the charge and its image charge are canceled each other and there is no radiation field anymore. Since the velocity in the charge is approximately constant before the collision, for the integral of Eq. (12),

$$\vec{n}\cdot\vec{r}(t)/c \simeq \vec{n}\cdot\vec{\beta}\,t$$

and

$$\int_{-\infty}^{0} (\vec{n}\times\vec{\beta})e^{i\omega(t-\vec{n}\cdot\vec{r}/c)}\,dt \simeq (\vec{n}\times\vec{\beta}) \int_{-\infty}^{0} e^{i\omega\left(1-\vec{n}\cdot\vec{\beta}\right)t}\,dt = \frac{\vec{n}\times\vec{\beta}}{\omega(1-\vec{n}\cdot\vec{\beta})}\; e^{i\omega\left(1-\vec{n}\cdot\vec{\beta}\right)t}\Big|_{-\infty}^{0}$$

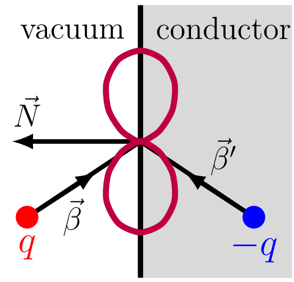
where $e^{i\omega\left(1-\vec{n}\cdot\vec{\beta}\right)t}$ is not defined at $t = -\infty$ which is far away from the conductor surface and shouldn't contribute to the radiation due to the interaction between the charge and a conductor surface. In real experiment, there should be some damping effect that will kill all terms at infinity. We therefore drop the term of $e^{-i\infty}$ and the energy spectrum is then

$$\frac{d^2E}{d\omega d\Omega} = \frac{q^2}{4\pi^2 c}\left|\frac{\vec{n}\times\vec{\beta}}{(1-\vec{n}\cdot\vec{\beta}\,)} - \frac{\vec{n}\times\vec{\beta}\,'}{(1-\vec{n}\cdot\vec{\beta}\,')}\right|^2 \tag{13}$$

The frequency spectrum of the transition radiation is therefore independent of frequency ω. Note that this is only true for good conductors. To further the calculation, we study two limit cases.

(a) $\beta << 1$

$$\frac{d^2E}{d\omega d\Omega} = \frac{q^2}{4\pi^2 c}\,[\vec{n}\times(\vec{\beta}-\vec{\beta}\,')]^2$$



Let $\vec{N}$ be unit vector normal to the surface at the collision point.

$$\vec{\beta}-\vec{\beta}\,' = 2(\,\vec{N}\cdot\vec{\beta}\,)\vec{N}$$

and

$$\frac{d^2E}{d\omega d\Omega} = \frac{q^2}{2\pi^2 c}(\vec{N}\cdot\vec{\beta}\,)^2\,|\vec{n}\times\vec{N}|^2 = \frac{q^2\beta_\perp^2}{2\pi^2 c}\,|\vec{n}\times\vec{N}|^2$$

where $\beta_\perp = \vec{\beta}\cdot\vec{N}$ is the velocity component normal to the surface and the radiation is independent of the tangential components of the velocity when a charge collides on a conductor.

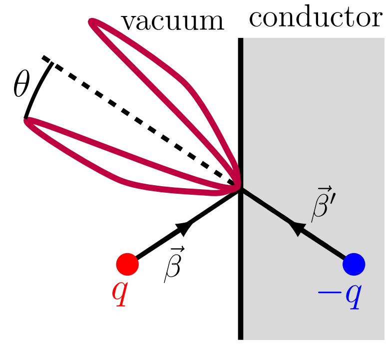
(b) $\beta \sim 1$

In the radiation energy formulas in Eq. (13), the first term with the denominator of $(1-\vec{n}\cdot\vec{\beta}\,)$ and the second term with the denominator of $(1-\vec{n}\cdot\vec{\beta}\,')$ dominates in a small cone along the forward direction of $\vec{\beta}$ and $\vec{\beta}\,'$, respectively. Since $\vec{\beta}$ of the incident charge points into the conductor, the radiation cone of the first term, which is the radiation from the incident charge, is directed into the conductor and an observer outside the conductor cannot see this radiation. The radiation cone of the second term, which is the radiation from the image charge beams toward outside of the conductor and can be observed from outside the conductor. For an observer outside the conductor, therefore,

$$\frac{d^2E}{d\omega d\Omega} = \frac{q^2}{4\pi^2 c}\,\frac{|\vec{n}\times\vec{\beta}\,'|^2}{(1-\vec{n}\cdot\vec{\beta}\,')^2} \tag{14}$$

Let θ be the angle between $\vec{n}$ and $\vec{\beta}'$. In Eq. (14)

$$\begin{aligned} 1-\vec{n}\cdot\vec{\beta}\,' &= 1-\beta\cos\theta \simeq 1-\left(1-\frac{1}{2\gamma^2}\right)\left(1-\frac{1}{2}\theta^2\right) \\ &\simeq \frac{1}{2\gamma^2}(1+\gamma^2\theta^2) \\ \vec{n}\times\vec{\beta}\,' &= \beta\sin\theta\simeq\theta \end{aligned}$$



$\theta_{max} = 1/\gamma$

Therefore

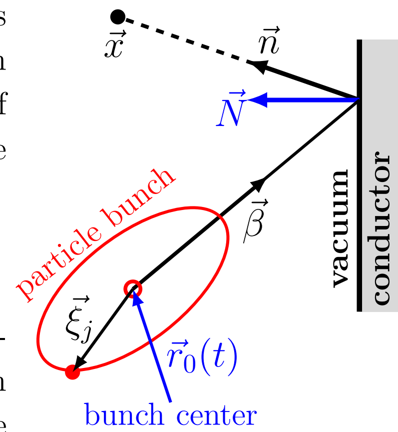
$$\frac{d^2E}{d\omega d\Omega} = \left(\frac{q^2\gamma^2}{\pi^2 c}\right)\frac{\gamma^2\theta^2}{(1+\gamma^2\theta^2)^2}$$

where the maximal radiation is at $\gamma^2\theta^2 = 1$ and $d^2E/d\omega d\Omega = 0$ at $\theta = 0$, *i.e.* the radiation cone has a hollow center, and the radiation intensity is proportional to γ^2.

(c) Coherent Transition Radiation

For a single charged particle collides with a conductor surface in the transition radiation, the radiation field has been calculated from Eq. (12),

$$\vec{C}(\omega) = \frac{i\omega q}{\sqrt{4\pi c}}\, e^{i\omega(\vec{x}\cdot\vec{n})/c} \int_{-\infty}^{0} dt \left\{ \left(\vec{n}\times\vec{\beta}\right) e^{i\omega[t-\vec{n}\cdot\vec{r}(t)/c]} - \left(\vec{n}\times\vec{\beta}\,'\right) e^{i\omega[t-\vec{n}\cdot\vec{r}\,'(t)/c]} \right\}$$



where $t = 0$ is chosen to be the time when the particle collides on the surface. For a bunch of charged particles collide with a conductor, we choose $t = 0$ as the time when the center of the bunch collides on the surface and the position of particle i can be written as

$$\vec{r}_j(t) = \vec{r}_0(t) + \vec{\xi}_j(t)$$

where $\vec{r}_0(t)$ is the position of the bunch center. If the observation distance $R = |\vec{x} - \vec{r}_0|$ is much larger than the bunch size, the difference of the radiation direction vector due to the difference in the collision point of different bunch particles is negligible, *i.e.*

$$\vec{n}_j = \frac{\vec{x}-\vec{r}_j}{|\vec{x}-\vec{r}_j|} \simeq \vec{n}$$

Moreover, the motion of the bunch particles relative to the bunch center is also negligible because $|\dot{\vec{\xi}}(t)| << \beta$, *i.e.*

$$\dot{\vec{r}}_j(t) \simeq \dot{\vec{r}}_0(t) = c\vec{\beta} \qquad \text{and} \qquad \vec{r}_j(t) \simeq c\vec{\beta}t + \vec{\xi}_j$$

The radiation field from a small bunch (bunch size $<< R$) of charged particles is then

$$\begin{aligned}
\vec{C}(\omega) &= \frac{i\omega q}{\sqrt{4\pi c}} \sum_j e^{i\omega(\vec{x}\cdot\vec{n}_j)/c} \int_{-\infty}^{t_j} dt \left\{ (\vec{n}_j\times\vec{\beta})e^{i\omega[t-\vec{n}_j\cdot\vec{r}_j(t)/c]} - (\vec{n}_j\times\vec{\beta}\,')e^{i\omega[t-\vec{n}_j\cdot\vec{r}_j\,'(t)/c]} \right\} \\
&\overset{\vec{n}_j\simeq\vec{n}}{=} \frac{i\omega q}{\sqrt{4\pi c}} e^{i\vec{k}\cdot\vec{x}} \sum_j \left[e^{-i\vec{k}\cdot\vec{\xi}_j}(\vec{n}\times\vec{\beta}\,) \int_{-\infty}^{t_j} e^{i(1-\vec{n}\cdot\vec{\beta})\omega t}\, dt - e^{-i\vec{k}\cdot\vec{\xi}_j'}(\vec{n}\times\vec{\beta}'\,) \int_{-\infty}^{t_j} e^{i(1-\vec{n}\cdot\vec{\beta}')\omega t}\, dt \right] \\
&= \frac{iq}{\sqrt{4\pi c}} e^{i\vec{k}\cdot\vec{x}} \sum_j \left[e^{-i\vec{k}\cdot\vec{\xi}_j} e^{i(1-\vec{n}\cdot\vec{\beta})\omega t_j} \frac{\vec{n}\times\vec{\beta}}{1-\vec{n}\cdot\vec{\beta}} - e^{-i\vec{k}\cdot\vec{\xi}_j'} e^{i(1-\vec{n}\cdot\vec{\beta}')\omega t_j} \frac{\vec{n}\times\vec{\beta}'}{1-\vec{n}\cdot\vec{\beta}'} \right]
\end{aligned}$$

where $\vec{k} = (\omega/c)\vec{n}$ and t_j is the collision time of the jth particle. Let $t_0 = 0$ be the collision time of the bunch center and $\vec{N}$ be the normal vector of the conductor surface. Then

$$t_j = \frac{\vec{r}_j\cdot\vec{N}}{\dot{\vec{r}}_j\cdot\vec{N}} = \frac{(\vec{r}_0+\vec{\xi}_j)\cdot\vec{N}}{c\vec{\beta}\cdot\vec{N}} = t_0 + \frac{\vec{\xi}_j\cdot\vec{N}}{c\vec{\beta}\cdot\vec{N}} = \frac{\vec{\xi}_j\cdot\vec{N}}{c\vec{\beta}\cdot\vec{N}}$$

where $\vec{r}_j \cdot \vec{N}$ is the distance of particle j to the conductor surface and $\dot{\vec{r}}_j \cdot \vec{N}$ is the speed of particle j approaching the surface. In the case of a relativistic beam, $\beta \sim 1$, the second term dominates the radiation into the outside of a conductor. For an observer outside the conductor, the radiation energy is then

$$\left(\frac{d^2E}{d\omega d\Omega}\right)_M = \frac{q^2}{4\pi^2 c}\left|\sum_j e^{-i\vec{k}\cdot\vec{\xi}'_j} e^{i(1-\vec{n}\cdot\vec{\beta}')\omega t_j}\frac{\vec{n}\times\vec{\beta}'}{1-\vec{n}\cdot\vec{\beta}'}\right|^2 = M^2 F(\vec{k})\left(\frac{d^2E}{d\omega d\Omega}\right)_1$$

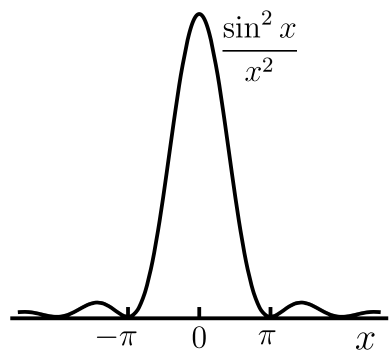
where M is the number of particles in a bunch and the form factor is

$$F(\vec{k}) = \left|\frac{1}{M}\sum_j e^{-i\vec{k}\cdot\vec{\xi}'_j} e^{i(1-\vec{n}\cdot\vec{\beta}')\omega t_j}\right|^2 \qquad \text{and} \qquad \left(\frac{d^2E}{d\omega d\Omega}\right)_1 = \frac{q^2}{4\pi^2 c}\frac{|\vec{n}\times\vec{\beta}'|^2}{(1-\vec{n}\cdot\vec{\beta}')^2}$$

For $\gamma >> 1$, the radiation is concentrated in the radiation cone with $(1-\vec{n}\cdot\vec{\beta}') \simeq 0$,

$$F(\vec{k}) = \left|\frac{1}{M}\sum_{j=1}^{M} e^{-i\vec{k}\cdot\vec{\xi}'_j}\right|^2$$

In the case of a uniform bunch of M particles with bunch length L and radius a, where $L >> a$,



$$\frac{1}{M}\sum_{j=1}^{M} e^{-i\vec{k}\cdot\vec{\xi}'_j} = \frac{1}{L}\int_{-L/2}^{L/2} e^{-ik\xi}d\xi = \frac{\sin(L\omega/2c)}{L\omega/2c}$$

and

$$\left(\frac{d^2E}{d\omega d\Omega}\right)_M = M^2\frac{\sin^2(L\omega/2c)}{(L\omega/2c)^2}\left(\frac{d^2E}{d\omega d\Omega}\right)_1$$

where $(d^2E/d\omega d\Omega)_1$ is independent of ω. The radiation is concentrated at low frequency. The cutoff frequency is at

$$L\omega/2c = \pi \qquad \Longrightarrow \qquad 1/\lambda_c = \omega_c/2\pi c = 1/L$$

The total coherent-radiation energy in this case is then

$$\begin{aligned}\left(\frac{dE}{d\Omega}\right)_M &= M^2\left(\frac{d^2E}{d\omega d\Omega}\right)_1\int_{-\infty}^{\infty}\frac{\sin^2(L\omega/2c)}{(L\omega/2c)^2}d\omega = \frac{2cM^2}{L}\left(\frac{d^2E}{d\omega d\Omega}\right)_1\int_{-\infty}^{\infty}\frac{\sin^2 x}{x^2}dx \\ &= \frac{2\pi c M^2}{L}\left(\frac{d^2E}{d\omega d\Omega}\right)_1 \end{aligned} \tag{15}$$

where

$$\int_{-\infty}^{\infty}\frac{\sin^2 x}{x^2}dx = \int_{-\infty}^{\infty}\frac{2\sin x\cos x}{x}dx = \int_{-\infty}^{\infty}\frac{\sin 2x}{x}dx = \pi$$

The total radiation intensity is proportional to M^2, the square of particle number in a bunch. It is therefore a coherent radiation.

17.3 Cherenkov Radiation (Section 13.4 in Jackson's Book)

Cherenkov radiation is when a charged particle travels through a medium at a velocity greater than the speed of light in that medium. When a fast electron passes through a liquid, for example, the radiation emitted by the electron has the appearance of a blue glow. (1958 Nobel Prize, Charenkov, Frank, and Tamm)

Let us recall the Fourier transformation of radiation field $\vec{B}$,

$$\vec{C}(\omega) = \frac{i\omega q}{\sqrt{4\pi c}}\, e^{i\omega(\vec{x}\cdot\vec{n})/c} \int_{-\infty}^{\infty} (\vec{n}\times\vec{\beta}\,)\, e^{i\omega[t-\vec{n}\cdot\vec{r}(t)/c]}\, dt \tag{16}$$

In this integral, it seems to be that a moving particle without acceleration can also radiate field. This is just a false impression. If $\vec{\beta}$ is constant, $\vec{n}\cdot\vec{r}(t)/c = \vec{n}\cdot\vec{\beta}\,t$ and

$$\int_{-\infty}^{\infty} (\vec{n}\times\vec{\beta}\,)\, e^{i\omega[t-\vec{n}\cdot\vec{r}(t)/c]} dt = (\vec{n}\times\vec{\beta}\,) \int_{-\infty}^{\infty} e^{i\omega(1-\vec{n}\cdot\vec{\beta}\,)t} dt = (\vec{n}\times\vec{\beta}\,)\,\frac{2\pi}{\omega}\,\delta(1-\vec{n}\cdot\vec{\beta}\,)$$

In vacuum, because $\vec{n}\cdot\vec{\beta} < 1$ and $\delta(1-\vec{n}\cdot\vec{\beta}\,) = 0$, there is no radiation from a moving charge with a constant velocity in vacuum. In a medium, however, it is a different story. As the radiation field travels with the speed of light in a medium, $v_\phi < c$, the retardation relation becomes

$$t = t_r + \frac{1}{v_\phi}\,[\vec{x} - \vec{r}(t_r)\,]\cdot\vec{n}$$

and the Fourier transformation of the radiation field in Eq. (16) should be changed as

$$\vec{C}(\omega) = \frac{i\omega q}{\sqrt{4\pi v_\phi}}\, e^{i\omega(\vec{x}\cdot\vec{n})/v_\phi} \int_{-\infty}^{\infty} (\vec{n}\times\vec{\beta}\,)\, e^{i\omega\left[t-\vec{n}\cdot\vec{r}(t)/v_\phi\right]}\, dt \tag{17}$$

For a particle with a constant velocity, $\vec{r}(t) = \vec{v}\,t$ and the oscillation phase in Eq. (17) is

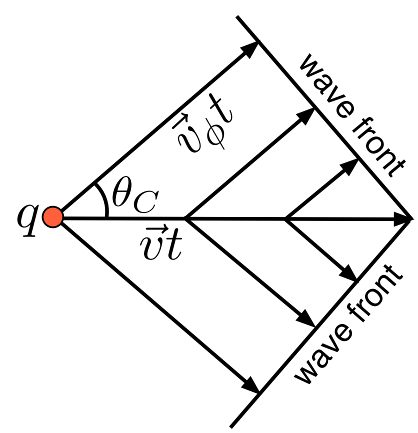
$$\omega\left[t - \frac{\vec{n}\cdot\vec{r}(t)}{v_\phi}\right] = \omega t\left(1 - \frac{\vec{n}\cdot\vec{v}}{v_\phi}\right)$$

This oscillation phase vanishes at a Cherenkov angle θ_C where

$$1 - \frac{\vec{n}\cdot\vec{v}}{v_\phi} = 1 - \frac{v}{v_\phi}\cos\theta_C = 0$$

or

$$\cos\theta_C = \frac{v_\phi}{v} \leq 1$$

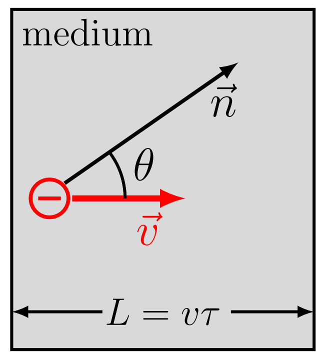


When a charged particle travels with a constant velocity faster than light in a medium, the integrand of the Fourier integral of the radiation field in Eq. (17) does not oscillate along the direction of $\vec{n}\cdot\vec{v} = v\cos\theta_C$ and the integral in Eq. (17) is thus no longer zero, *i.e* the charged particle emits a radiation. This Cherenkov radiation travels only in the direction along the surface of a cone around particle trajectory. The angle of the core is $2\theta_C$, *i.e.* the angle between the direction of the radiation and the direction of the particle motion is θ_C. Since the speed of light in a medium depends usually on the frequency of light as

$$v_\phi = \frac{c}{n(\omega)} \qquad \text{and} \qquad \theta_C = \cos^{-1}\frac{1}{\beta n(\omega)}$$

where $n(\omega)$ is the index of reflection of a medium, which could be functions of frequency of the radiation. The Cherenkov angle θ_C is usually a function of the radiation frequency and the radiation of different frequency travels in different direction, which results in a dispersion of the radiation.

(a) Frequency Spectrum of Energy in Solid Angle of Cherenkov Radiation



To calculate the energy spectrum of the Cherenkov radiation, we consider a charged particle with a constant velocity passing through a medium of finite thickness during a time τ. The thickness of the medium is $L = v\tau$, where v is the constant velocity of the particle. Outside the medium, the particle travels slower than the speed of light and the integrand of the radiation field in Eq. (16) oscillates and the integral cancels to zero in all the observation direction. When the particle passes through the medium, the energy spectrum of the Cherenkov radiation is

$$\frac{d^2E}{d\omega d\Omega} = \frac{1}{\pi}|\vec{C}(\omega)|^2 = \frac{q^2\omega^2}{4\pi^2c^2v_\phi}\left|\int_{-\tau/2}^{\tau/2}(\vec{n}\times\vec{v}\,)\,e^{i\omega\left(t-\vec{n}\cdot\vec{r}/v_\phi\right)}\,dt\right|^2 \tag{18}$$

where $(\vec{n}\times\vec{v}\,) = v\sin\theta$ is independent of t, $\vec{r}(t) = \vec{v}\,t$, and

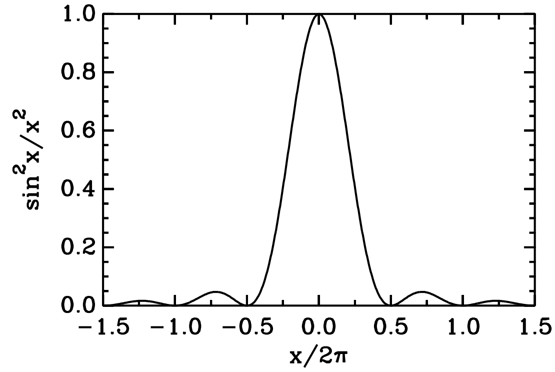
$$(t - \vec{n}\cdot\vec{r}/v_\phi) = t\left(1 - \frac{v}{v_\phi}\cos\theta\right) = t\left(1 - \frac{\cos\theta}{\cos\theta_C}\right)$$

Thus,

$$\begin{aligned}\frac{d^2E}{d\omega d\Omega} &= \frac{q^2\omega^2}{4\pi^2c^2v_\phi}\,v^2\sin^2\theta\left|\int_{-\tau/2}^{\tau/2}e^{i\omega t\left[1-(v/v_\phi)\cos\theta\right]}\,dt\right|^2 \\ &= \frac{q^2v^2}{2\pi^2c^2v_\phi}\left(\omega\tau\sin^2\theta\right)\frac{\sin^2\left(\omega\tau\left[1-(v/v_\phi)\cos\theta\right]/2\right)}{\omega\tau\left[1-(v/v_\phi)\cos\theta\right]^2/2} \\ &= \frac{q^2v^2}{2\pi^2c^2v_\phi}\left[(\omega\tau)^2 - (\omega\tau - 2x)^2\cos^2\theta_C\right]\frac{\sin^2 x}{x^2}\end{aligned} \tag{19}$$

where $x = \omega\tau\left[1-(v/v_\phi)\cos\theta\right]/2$. Note that the calculation of the radiation energy in Eq. (19) does not explicitly require the condition of $v > v_\phi$. Due to a finite thickness of the medium, the radiation is nonzero even when $v < v_\phi$. The radiation energy is, however, concentrated in a peak centered at an observation angle that satisfies the Cherenkov condition,

$$\frac{v}{v_\phi(\omega)}\cos\theta = 1$$



$x = \omega\tau\left[1-(v/v_\phi)\cos\theta\right]/2$

where the speed of light in medium $v_\phi(\omega)$ could be a function of radiation frequency (dispersion). The condition of $v > v_\phi$ is therefore required for the radiation to attend its maximum intensity at $x \in [-\pi,\pi]$. When $v < v_\phi$, $[1-(v/v_\phi)\cos\theta] > 0$ and $x > \pi$ for sufficiently large $\omega\tau = \omega L/v$. Therefore, for a given frequency and with a sufficiently thick medium, the radiation energy is away from the central peak and the radiation is insignificant when $v < v_\phi$. For a sufficiently small $\omega\tau$, on the other hand, one could have $0 < x < \pi$ and the radiation is still significant even when $v < v_\phi$. Note that, in Eq. (19),

$$\lim_{(\omega\tau/2)\to\infty}\frac{\sin^2\left(\omega\tau\left[1-(v/v_\phi)\cos\theta\right]/2\right)}{\omega\tau\left[1-(v/v_\phi)\cos\theta\right]^2/2} \;=\; \pi\,\delta\left(1-\frac{v}{v_\phi}\cos\theta\right)$$

In the high frequency range of $\omega >> v/L$, therefore,

$$\frac{d^2E}{d\omega d\Omega} = \frac{q^2Lv}{2\pi c^2 v_\phi}\,\omega\,\left(1-\cos^2\theta\right)\,\delta\left(1-\frac{v}{v_\phi}\cos\theta\right) \tag{20}$$

The peak of the radiation spectrum becomes narrower when the medium is thicker and the radiation energy increases linearly with the thickness of the medium.

(b) Frequency Spectrum of Total Radiation Energy of Cherenkov Radiation

For a charged particle traveling through a medium of thickness $L = v\tau$, the total radiation energy in unit frequency interval can be calculated from Eq. (19) as

$$\begin{aligned}\frac{dE}{d\omega} &= \frac{q^2v^2}{\pi^2c^2v_\phi}(2\pi)\int_0^\pi \sin\theta d\theta\,\frac{\sin^2\theta}{\left[1-(v/v_\phi)\cos\theta\right]^2}\sin^2\left[\frac{\omega\tau}{2}\left(1-\frac{v}{v_\phi}\cos\theta\right)\right]\\ &= \frac{2q^2v^2}{\pi c^2 v_\phi}\int_{-1}^{1}\frac{1-\cos^2\theta}{\left[1-(v/v_\phi)\cos\theta\right]^2}\sin^2\left[\frac{\omega\tau}{2}\left(1-\frac{v}{v_\phi}\cos\theta\right)\right]\,d\cos\theta\end{aligned}$$

Let

$$x = \frac{\omega\tau}{2}\left(1-\frac{v}{v_\phi}\cos\theta\right)\,, \quad \cos\theta = \frac{v_\phi}{v}\left(1-\frac{2}{\omega\tau}x\right)\,, \quad d\cos\theta = -\left(\frac{2v_\phi}{v\omega\tau}\right)\,dx$$

Note that $(d^2E/d\omega d\Omega)$ in Eq. (19) is significantly nonzero only when $x \in [-\pi, \pi]$. Thus

$$
\begin{aligned}
\frac{dE}{d\omega} &= \frac{2q^2v^2}{\pi c^2 v_\phi}\left(\frac{\omega\tau}{2}\right)^2\left(\frac{2v_\phi}{v\omega\tau}\right)\left(\frac{v_\phi}{v}\right)^2 \int\limits_{\omega\tau(1-v/v_\phi)/2}^{\omega\tau(1+v/v_\phi)/2} \frac{\sin^2 x}{x^2}\left[\left(\frac{v}{v_\phi}\right)^2 - \left(1-\frac{2x}{\omega\tau}\right)^2\right] dx \\
&\simeq \frac{q^2\omega\tau v_\phi^2}{\pi c^2 v}\,\frac{4}{\omega^2\tau^2}\int\limits_{-\pi}^{\pi}\left[\frac{\omega^2\tau^2(v^2-v_\phi^2)}{4v_\phi^2\,x^2} + \frac{\omega\tau}{x} - 1\right]\sin^2 x\,dx \\
&= \frac{2q^2}{c^2 v\omega\tau}\left[\omega^2\tau^2\left(v^2-v_\phi^2\right) - 2v_\phi^2\right] \\
&= \frac{2q^2}{c}\left(\frac{\omega L}{c}\right)\left\{1 - \frac{1}{n^2}\left[\frac{1}{\beta^2} + 2\left(\frac{c}{\omega L}\right)^2\right]\right\}
\end{aligned}
\tag{21}
$$

where $v_\phi = c/n$ and n is the index of refraction of the medium that usually depends on the frequency, and

$$
\int_{-\pi}^{\pi}\frac{\sin^2 x}{x^2}dx \simeq 2\int_0^\infty \frac{\sin^2 x}{x^2}dx = 2\pi \qquad \text{and} \qquad \int_{-\pi}^{\pi}\frac{\sin^2 x}{x}dx = 0
$$

The requirement of $dE/d\omega > 0$ in Eq. (21) yields the condition for the lowest (cutoff) frequency of the Cherenkov radiation,

$$
\frac{dE}{d\omega} > 0 \qquad \Longrightarrow \qquad n^2 > \frac{1}{\beta^2} + 2\left(\frac{c}{\omega L}\right)^2
$$

For a given particle speed v, the lowest (cutoff) frequency of the Cherenkov radiation is therefore at

$$
\frac{2}{(\omega_{min}L)^2} = \frac{1}{v_\phi^2} - \frac{1}{v^2}
$$

Note that $dE/d\omega$ in Eq. (21) increases linearly with radiation frequency ω when $\omega >> c/L$. The visible spectrum of the Cherenkov radiation is therefore dominated by blue light or blue glow. This is the reason that it was mistakenly identified as a fluorescent before Cherenkov. At the high frequency,

$$
\frac{c}{2q^2}\left(\frac{dE}{d\omega}\right) \simeq \left(1 - \frac{v_\phi^2}{v^2}\right)\left(\frac{\omega L}{c}\right) \qquad \text{for} \quad \omega >> c/L \tag{22}
$$

The slop of the linear $(\omega L/c)$-dependence of the radiation spectrum of the Cherenkov radiation at high frequency is proportional to $(1 - v_\phi^2/v^2)$.

(c) Number of Photons in Cherenkov Radiation

Since the energy quanta in the radiation with frequency ω is $\hbar\omega$, the number of photons in the Cherenkov radiation in the unit frequency interval can be calculated from Eq. (21) as

$$
\frac{dN}{d\omega} = \frac{1}{\hbar\omega}\frac{dE}{d\omega} = \frac{2q^2L}{\hbar c^2}\left(1 - \frac{v_\phi^2}{v^2} - \frac{2v_\phi^2}{L^2\omega^2}\right)
$$

If the dispersion of the medium is negligible, v_ϕ is approximately independent of ω, and the total number of photons in a frequency range of $[\omega_1, \omega_2]$ is then

$$N = \int_{\omega_1}^{\omega_2} \frac{dN}{d\omega}\, d\omega = \frac{2q^2 L}{\hbar c^2} \left[(\omega_2 - \omega_1) \left(1 - \frac{v_\phi^2}{v^2} \right) - \frac{2v_\phi^2}{L^2} \left(\frac{1}{\omega_1} - \frac{1}{\omega_2} \right) \right] \tag{23}$$

where $L = v\tau$ is the thickness of the medium.

(d) Effect of Dispersion to Cherenkov Radiation

The Cherenkov radiation is peaked at the observation angle of $\cos\theta = v_\phi/v$. If there is a significant dispersion in the medium, the radiation peaks with different radiation frequencies are spread out with different observation angles. If $\min\{v_\phi(\omega)\}$ and $\max\{v_\phi(\omega)\}$ are the minimal and maximal of the speed of light due to the dispersion, respectively, the smallest and largest observation angle of the Cherenkov radiation are

$$\cos\theta_{min} = \frac{\min\{v_\phi(\omega)\}}{v} \qquad \text{and} \qquad \cos\theta_{max} = \frac{\max\{v_\phi(\omega)\}}{v}$$

For a weak frequency dependence of $v_\phi(\omega)$, the difference between θ_{max} and θ_{min} is small, and the Cherenkov radiation could emit over a narrow band of different color with high frequency radiation dominates in intensity. For a weak frequency dependence of $v_\phi(\omega)$, in Eq. (21), we can approximate

$$v_\phi(\omega) \simeq v_\phi(\omega_0) + v'_\phi(\omega_0)(\omega - \omega_0)$$

where $v_\phi(\omega_0) = [\min(v_\phi) + \max(v_\phi)]/2$ and ω_0 is the frequency ω for $v_\phi(\omega) = v_\phi(\omega_0)$.

17.4 Channeling Radiation of Relativistic Electrons in Crystal

1. Relativistic Beam Electrons Passing Through a Crystal

Considering a relativistic electron beam passing through a crystal lattice with the longitudinal velocity of beam electrons parallel to a crystal direction $[hkl]$, two effects could prevent the electron passing through the crystal without being perturbed, interference of the electron by the lattice and scattering of the electron by the crystal ions.

Interference of Electron by a Crystal

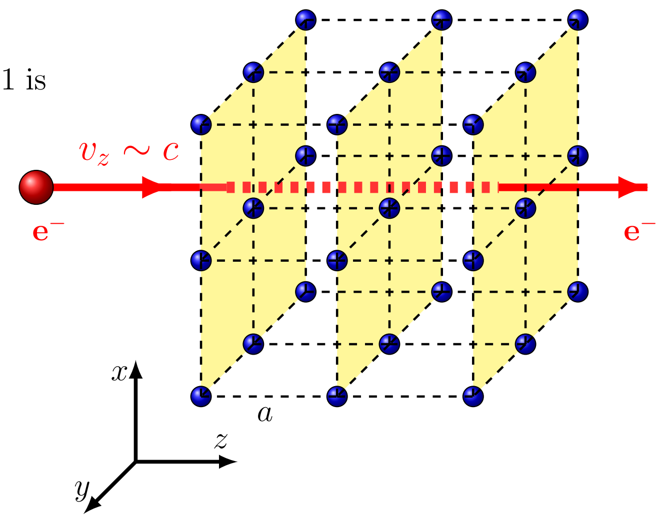
The wavelength of an electron with $\beta \sim 1$ is

$$\lambda = \frac{h}{p} \simeq \frac{hc}{E} \sim \frac{10^{-6}}{E \text{ in eV}}$$

and the constructive interference occurs at

$$a \sin\theta = n\lambda\,, \qquad \text{where} \;\; a \gtrsim 10^{-8}$$

E	λ	effect from crystal
KeV	10^{-9}	interference
MeV	10^{-12}	channeling



Scattering of Electron by Crystal Ions

From Eq. (20) of Section 15.2, the energy loss of an electron due to a soft collision with an ion at $\gamma >> 1$ is

$$\frac{dE}{d\Omega} \sim \left(\frac{\gamma r_0}{b}\right)^2 \left(\frac{Qq}{b}\right) << \gamma mc^2 \tag{24}$$

where $(Qq/b) << \gamma mc^2$ is the maximal electric potential energy of the incident electron which is much smaller than the kinetic energy of the electron, $r_0 \sim 3 \times 10^{-15}$ m is the classical electron radius and $r_0 << b$. The longitudinal motion with $v_z \sim c$ of an electron is unperturbed as it passing through a crystal.

2. Non-Relativistic Transverse Motion of Beam Electrons

Considering an electron passing through a crystal with $v_z \sim c$ and $v_x << c$, $v_y << c$, the transverse motion (v_x and v_y) could be perturbed by the crystal lattice. To study the transverse motion of the electron inside a crystal, we go to the rest frame of the beam that moves with $v_z \sim c$ in the lab frame. For the interaction between the electron and ions in the crystal, in the lab frame, the electric and magnetic field from the ions of crystal is

$$\vec{E}' = -\frac{1}{e}\nabla V(\vec{r}) \qquad \text{and} \qquad \vec{B}' = 0 \tag{25}$$

where $V(\vec{r})/e$ is the electric potential from all the ions in the lattice in the lab frame. In the beam rest frame with $\vec{\beta} = \beta_z \vec{e}_z$,

$$\begin{aligned}
\vec{E} &= \gamma(\vec{E}\,' + \vec{\beta} \times \vec{B}\,') - \frac{\gamma^2}{\gamma+1} \vec{\beta}\,(\vec{\beta} \cdot \vec{E}\,') \;=\; \gamma \vec{E}\,' - \frac{\gamma^2}{\gamma+1} \beta_z^2 E_z'\, \vec{e}_z \qquad (26)\\
&\overset{\gamma >> 1}{=} \gamma E_x'\, \vec{e}_x + \gamma E_y'\, \vec{e}_y + E_z'\, \vec{e}_z \;\overset{\gamma >> 1}{=}\; \gamma \vec{E}\,'_\perp \\
\vec{B} &= \gamma(\vec{B}\,' - \vec{\beta} \times \vec{E}\,') - \frac{\gamma^2}{\gamma+1} \vec{\beta}\,(\vec{\beta} \cdot \vec{B}\,') \;=\; -\gamma \beta_z (\vec{e}_z \times \vec{E}\,') \qquad (27)
\end{aligned}$$

where $E_z = E_z' << |\vec{E}_\perp| = \gamma |\vec{E}_\perp'|$, the z-component of the $\vec{E}$-field is not important and negligible. In the beam rest frame, the velocity of a beam electron is $\vec{v} = (v_x, v_y, 0)$ and non-relativistic. The Lorenz force from the crystal ions to the electron is

$$\begin{aligned}
\vec{F} \;=\; e\left(\vec{E} + \frac{\vec{v}}{c} \times \vec{B}\right) &= e\gamma \left[\vec{E}\,'_\perp - \beta_z \frac{\vec{v}}{c} \times (\vec{e}_z \times \vec{E}\,')\right] \\
&\overset{v << c}{=} e\gamma \vec{E}\,'_\perp \;=\; -\gamma \nabla V(x,y) \qquad (28)
\end{aligned}$$

where $|\vec{v}\,|/c << 1$. The Hamiltonian for the non-relativistic transverse motion of a beam electron inside a crystal in the beam rest frame is then

$$H = \frac{1}{2m}(p_x^2 + p_y^2) + \gamma V(x,y) \qquad (29)$$

where $V(x,y)$ is the projection of 3-dimensional lattice potential of the crystal into the beam transverse plane. Because of the periodicity of the crystal lattice, this 2-dimensional potential is periodic

$$V(x,y) = V(x + a_1, y) = V(x, y + a_2) \qquad (30)$$

where a_1 and a_2 are the lattice constants in the two-dimensional transverse plane. Since this is a non-relativistic electron motion in a potential well of $\gamma \nabla V(x,y)$, it should be solved from Shrödinger equation,

$$-\frac{\hbar^2}{2m} \nabla^2 \psi_n(x,y) + \gamma V(x,y) \psi_n(x,y) = E_n \psi_n(x,y) \qquad (31)$$

The eigenstates of the Shrödinger equation with a periodic potential well are the Bloch wavefunctions

$$\psi_n(x,y) = e^{i2\pi(\Omega_1 x/a_1 + \Omega_2 y/a_2)}\, u_{n,\vec{\Omega}}(x,y) \qquad (32)$$

where Ω_1 and Ω_2 are two constants that can have any values, with $\Omega_1 \in [0,1)$ and $\Omega_2 \in [0,1)$ being called the first Brillouin zone, and $u_{n,\vec{\Omega}}(x,y)$ is a periodic function with the period of the potential well,

$$u_{n,\vec{\Omega}}(x,y) \;=\; u_{n,\vec{\Omega}}(x + a_1, y) \;=\; u_{n,\vec{\Omega}}(x, y + a_2) \qquad (33)$$

For a potential well $\gamma V(x,y) < 0$, the eigenstates $\psi_n(x,y)$ are either bound states with $E_n(\vec{\Omega}\,) < 0$ or unbound states with $E_n(\vec{\Omega}\,) > 0$.

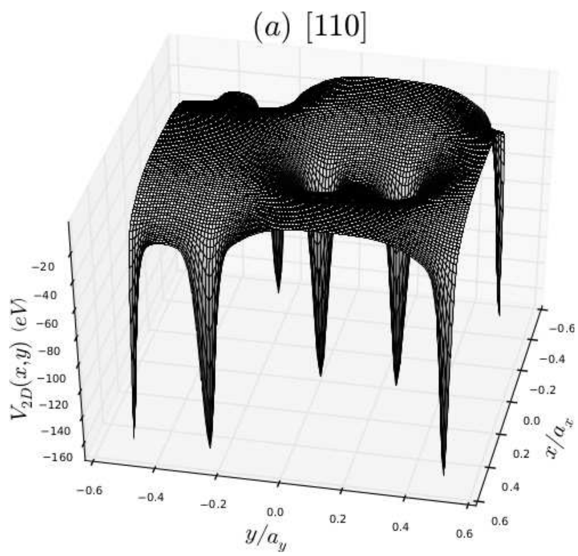
Bound Bloch Eigenstates with $E_n(\vec{\Omega}\,) < 0$

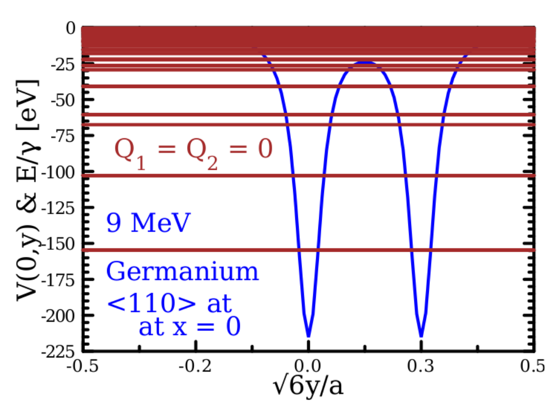
The bound eigenstates are discrete states with the electron transverse motion being confined inside the potential well, *i.e.* the transverse motion is localized in each unit cell of lattice and the transverse position of the electron is well defined (organized motion) during the electron passing through the crystal. The electrons in the bound states can therefore channel through a crystal without being scattered from lattice ions.

Unbound Bloch Eigenstates with $E_n(\vec{\Omega}\,) > 0$

The energy separations of the unbound states are diminished rapidly as $E_n(\vec{\Omega}\,)$ increases from zero and, therefore, the unbound Bloch states form a continue energy spectrum. The electrons in the unbound states are not localized, *i.e.* the density of the wavefunctions spread in whole lattice and becomes more uniform in the crystal as $E_n(\vec{\Omega}\,)$ increases. The transverse motion of the electrons in the unbound states can thus be scattered by lattice ions as the electrons randomly move transversely in the crystal.

Channeling Radiation Spectrum





The transitions between the bound Bloch eigenstates of the electron transverse motion result in emissions of photons. The frequency of the radiation observed in the beam rest frame is simply

$$\omega_{ij} = \frac{1}{\hbar}(E_i - E_j) \tag{34}$$

where $\hbar = 6.6 \times 10^{-16} eV \cdot s$. In the lab frame, the radiation frequency observed is shifted due to the relativistic Doppler effect as

$$\omega = \frac{\omega_{ij}}{\gamma(1-\beta\cos\theta)} \tag{35}$$

where θ is the angle between the beam direction $\hat{\beta} = \vec{e}_z$ and the direction from the beam to the observer. If the observation is made from the forward direction of the beam, in the front radiation cone, $\theta \simeq 0$,

$$\omega \;=\; \frac{\omega_{ij}}{\gamma(1-\beta)} \;\overset{\gamma >> 1}{\cong}\; 2\gamma\omega_{ij} \tag{36}$$

For $(E_2 - E_1) \sim 50\gamma eV$ and $\gamma = 18$ in the calculated energy levels in the figure for a 9 MeV electron beam passing through Germanium crystal, $\omega/2\pi \sim 10^{19}$ (32 KeV), which is in the range of hard X-ray (1 to 100 KeV).

HW
13.9 on Jackson Book

Extra 17.1 For the transition radiation, show that for $\beta << 1$,

$$\frac{dE}{d\omega} \sim (\vec{N} \cdot \vec{\beta})^2$$

and for $\beta \sim 1$,

$$\frac{dE}{d\omega} \sim \ln \gamma$$

Extra 17.2 Consider N uniformly distributed relativistic electrons moving around a circular orbit with radius R and circular frequency ω_0.
(a) Calculate the frequency spectrum of the radiation from individual electron.
(b) Calculate the form factor of N electrons.

Extra 17.3 Consider a bunch of relativistic electrons of length L is incident to and absorbed by a conducting surface, Show that the coherent transition radiation is

$$\frac{dE}{d\Omega} \sim \frac{1}{L}$$

Extra 17.4 In the argument of Cherenkov radiation of a charged particle moving in a medium with a negligible acceleration, we use the Fourier transformation of the radiation field in Eq. (16) to demonstrate the none-zero Fourier transformation of the radiation field occurs when the particle moves faster than the speed of light v_ϕ in the medium and when the radiation direction $\vec{n}$ satisfies $\vec{n} \cdot \vec{v} = v_\phi$, where $\vec{v}$ is the particle velocity.

Demonstrate the existence of Cherenkov radiation using the equation of the radiation field

$$\vec{E}_{rad}(\vec{x}, t) = q \left. \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{c\, R\, (1 - \vec{n} \cdot \vec{\beta})^3} \right|_{retard}$$

with a negligible acceleration $|\dot{\vec{\beta}}| << 1$. Note that the acceleration of a charged particle inside a medium is

$$\frac{d\vec{p}}{dt} = -\nabla V \qquad \Longrightarrow \qquad \dot{\vec{\beta}} = -\frac{1}{\gamma mc} \nabla V$$

where V is the interaction potential between the charged particle and the charge-polarized molecules inside the medium. With a relativistic particle, $\gamma >> 1$ and $|\nabla V|/mc << 1$, $|\dot{\vec{\beta}}| << 1$.